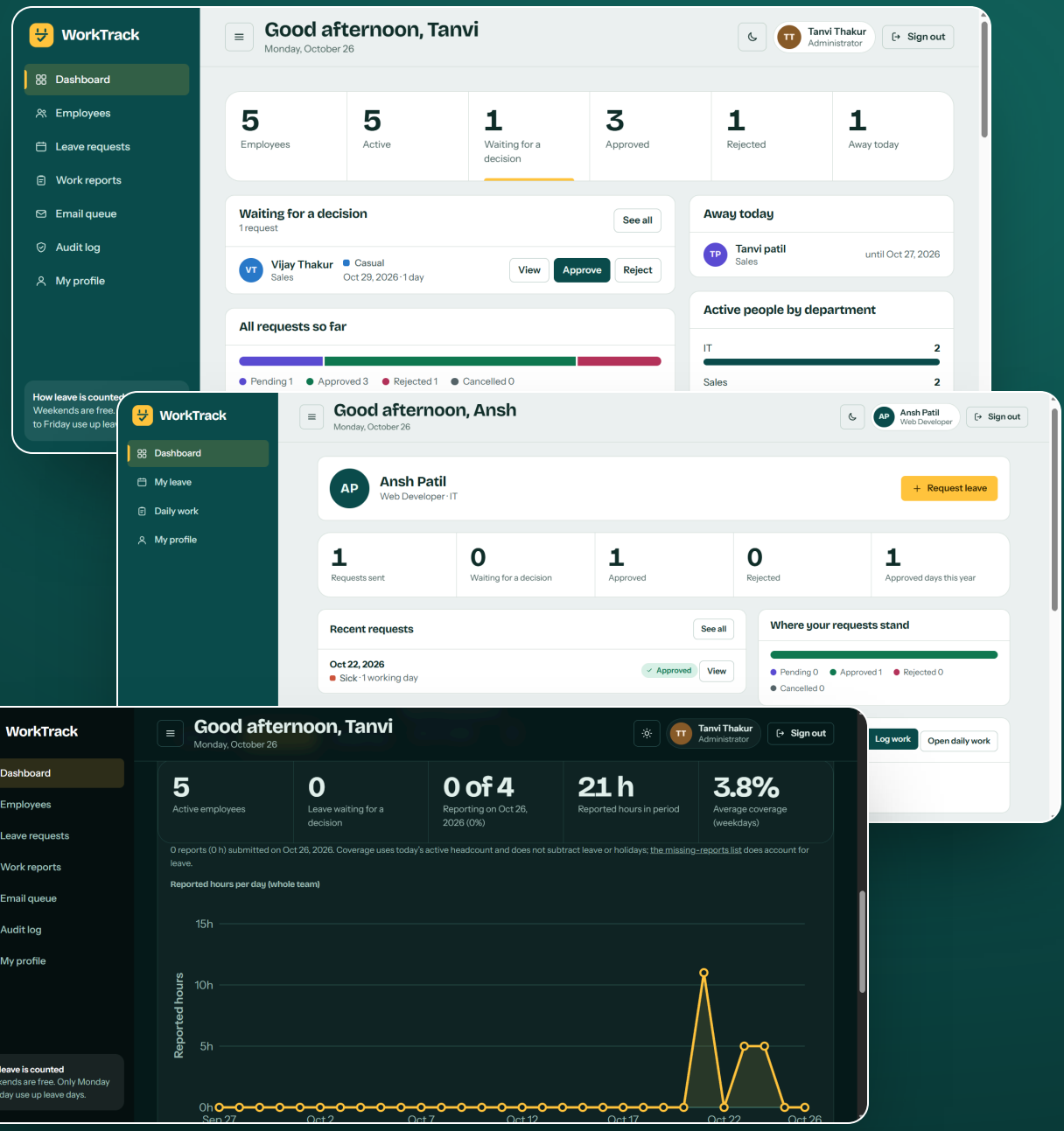
Personal full-stack project

# WorkTrack

## Employee and Work Management System

A full-stack web application designed to manage employee information, daily work activities, leave requests, and administrative workflows.

Java | Spring Boot | MySQL | JavaScript | HTML | CSS




Personal learning project. Runs locally; not publicly deployed.

# What WorkTrack does



WorkTrack brings three everyday workplace tasks into one application: keeping an employee directory, recording what people worked on each day, and handling leave requests from submission to decision. Two roles, **Administrator** and **Employee**, see different pages and have different permissions.

## Employee management

Administrators add, edit and deactivate employees. New people receive an invitation email to set their own password.

## Daily work tracking

Employees log tasks with hours and status. Administrators review team reports and see who has not reported.

## Leave management

Employees request leave; administrators approve or reject it. Working days are counted automatically and weekends are free.

## Dashboards and reports

Summary cards and charts for administrators and employees, with CSV and Excel export of work reports.

## Email notifications

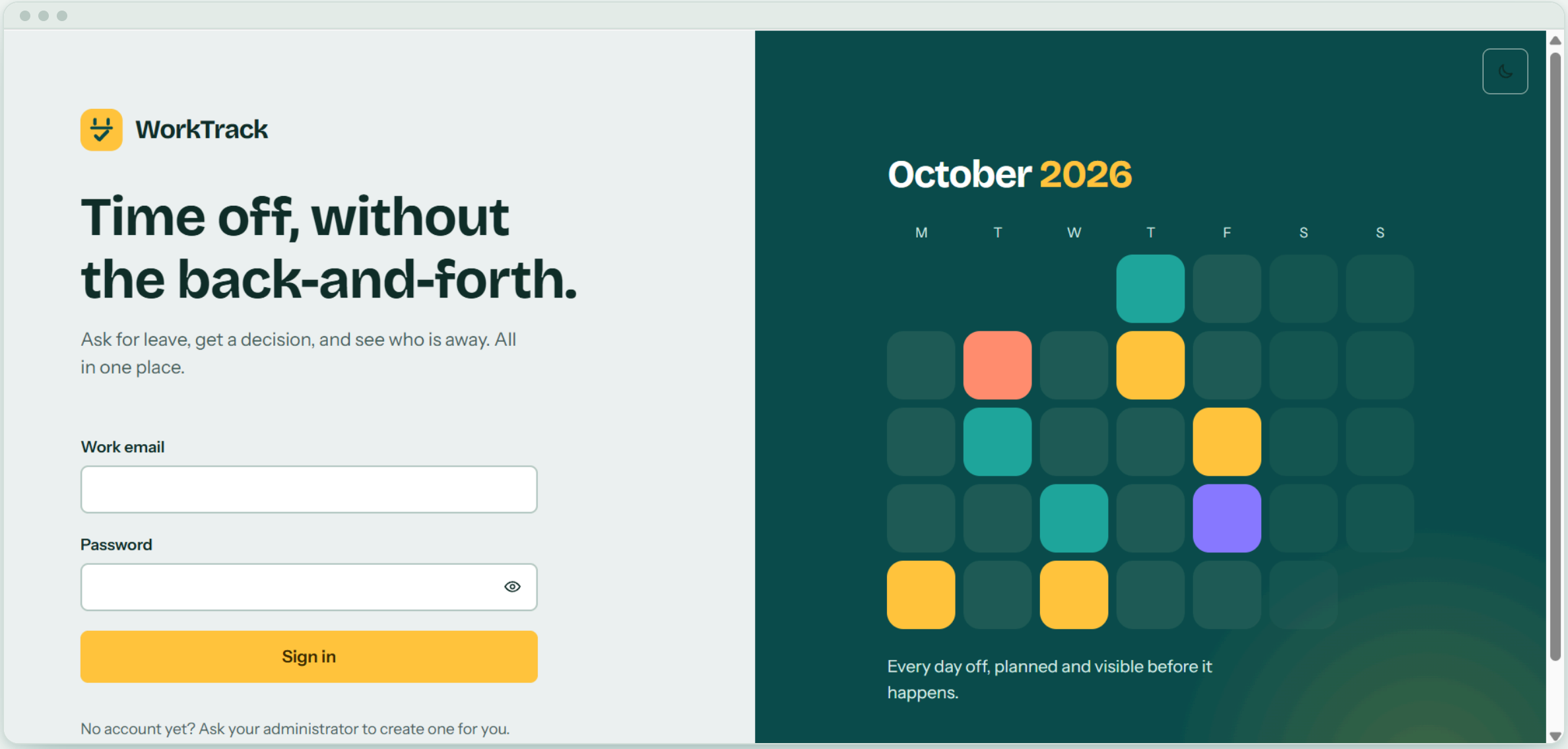
Invitations, leave decisions and account changes are emailed, with a queue that shows each message's status.

## Access and history

Role-based access, password change, and an audit log of who did what and when.

# Sign in

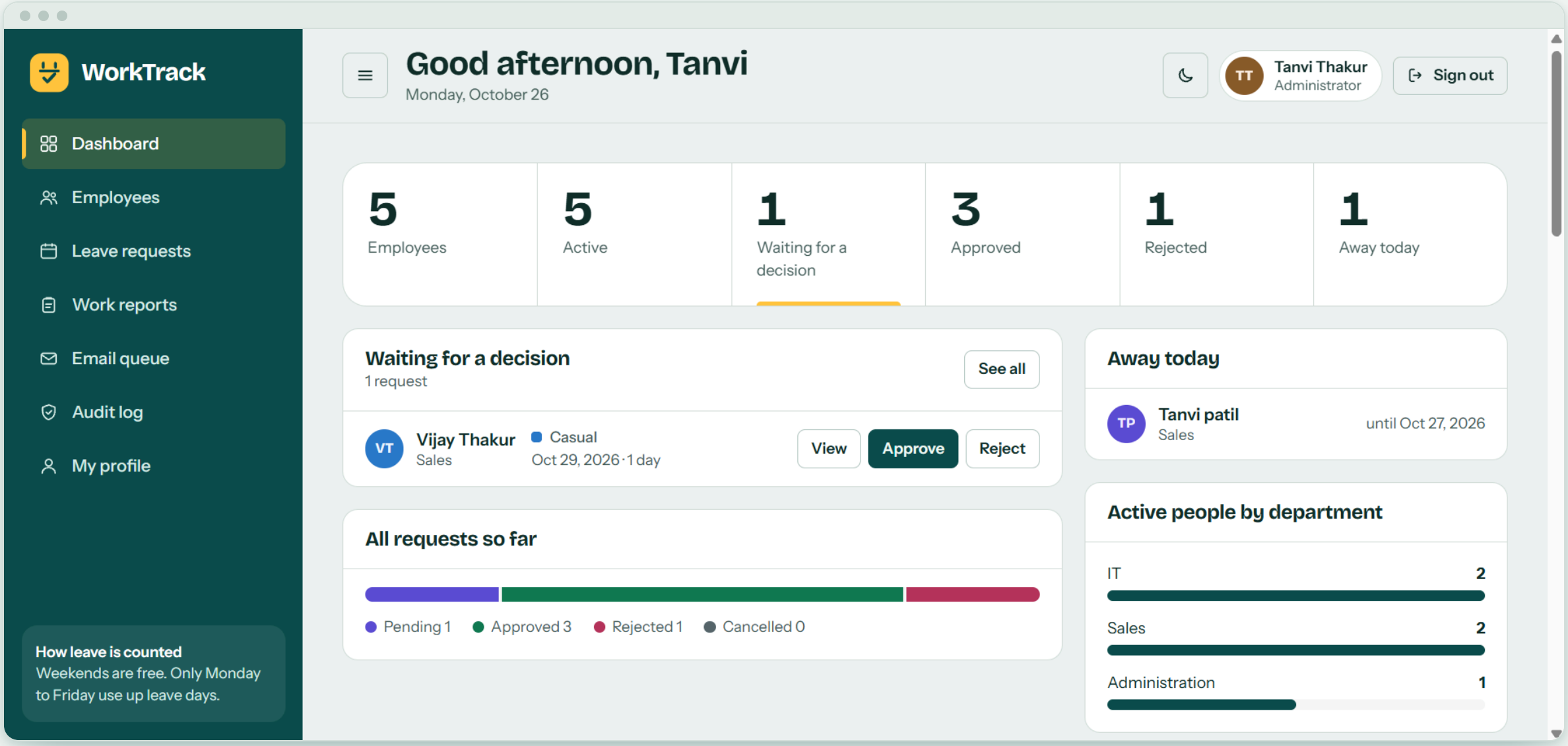





**Secure sign-in.** Users sign in with a work email and password. The eye icon shows or hides the password, and a toggle in the corner switches between light and dark themes.

# Administrator dashboard

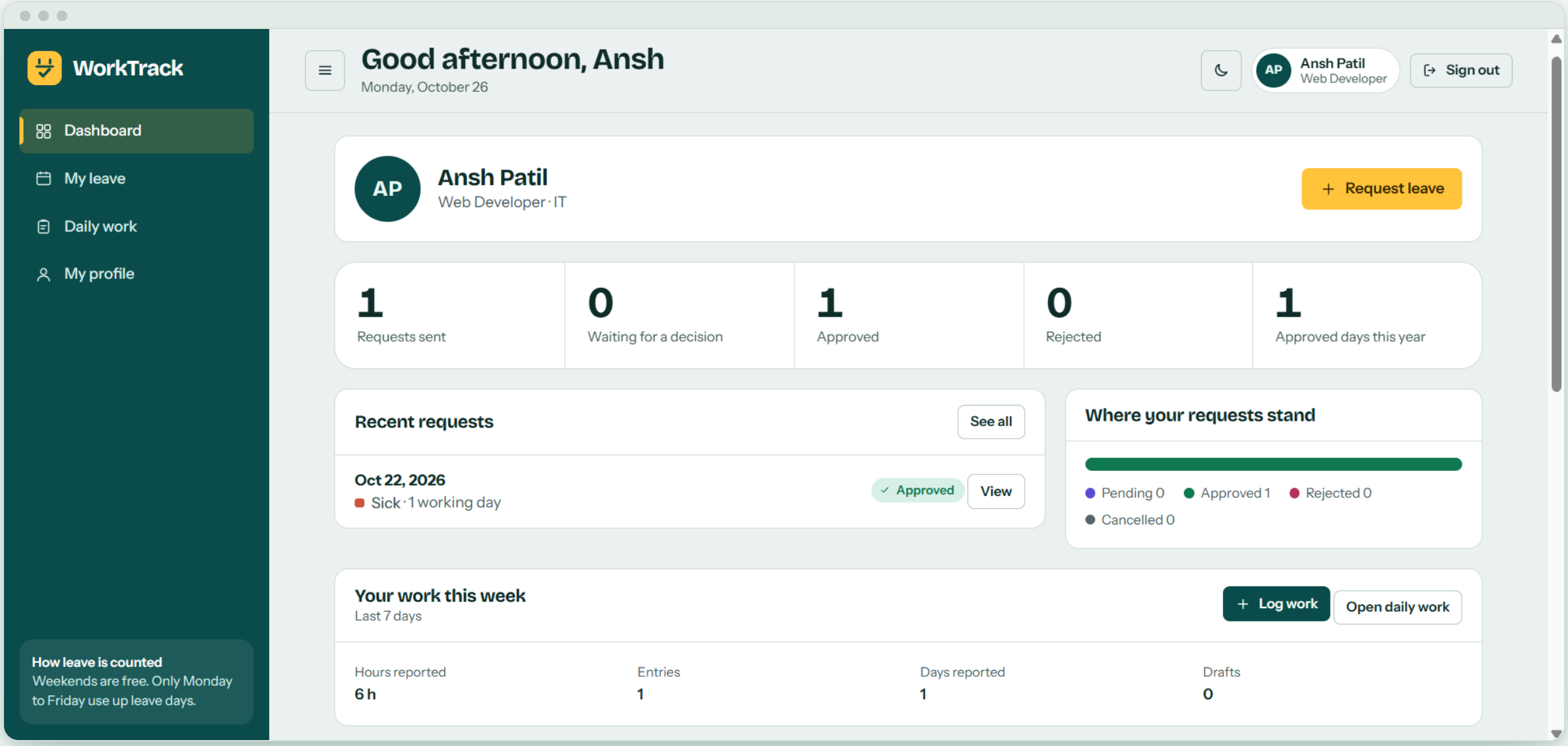





**Everything an admin needs at a glance.** Employee counts, leave waiting for a decision (approve or reject from here), who is away today, and active people by department.

# Employee dashboard

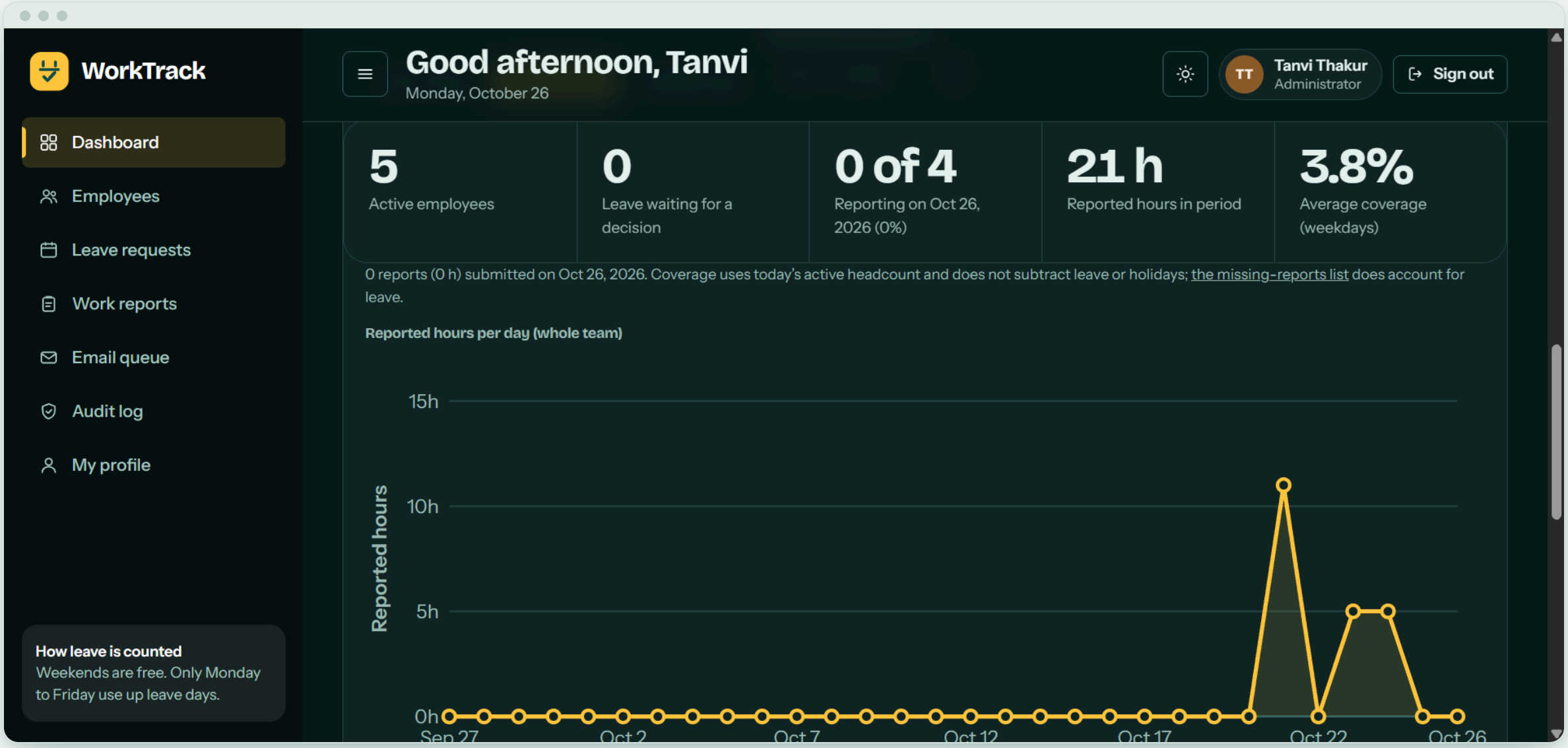





**A personal view for each employee.** Leave summary, recent requests with their status, and a snapshot of work reported this week, with shortcuts to request leave or log work. The sidebar shows employee pages only.

# Light and dark themes

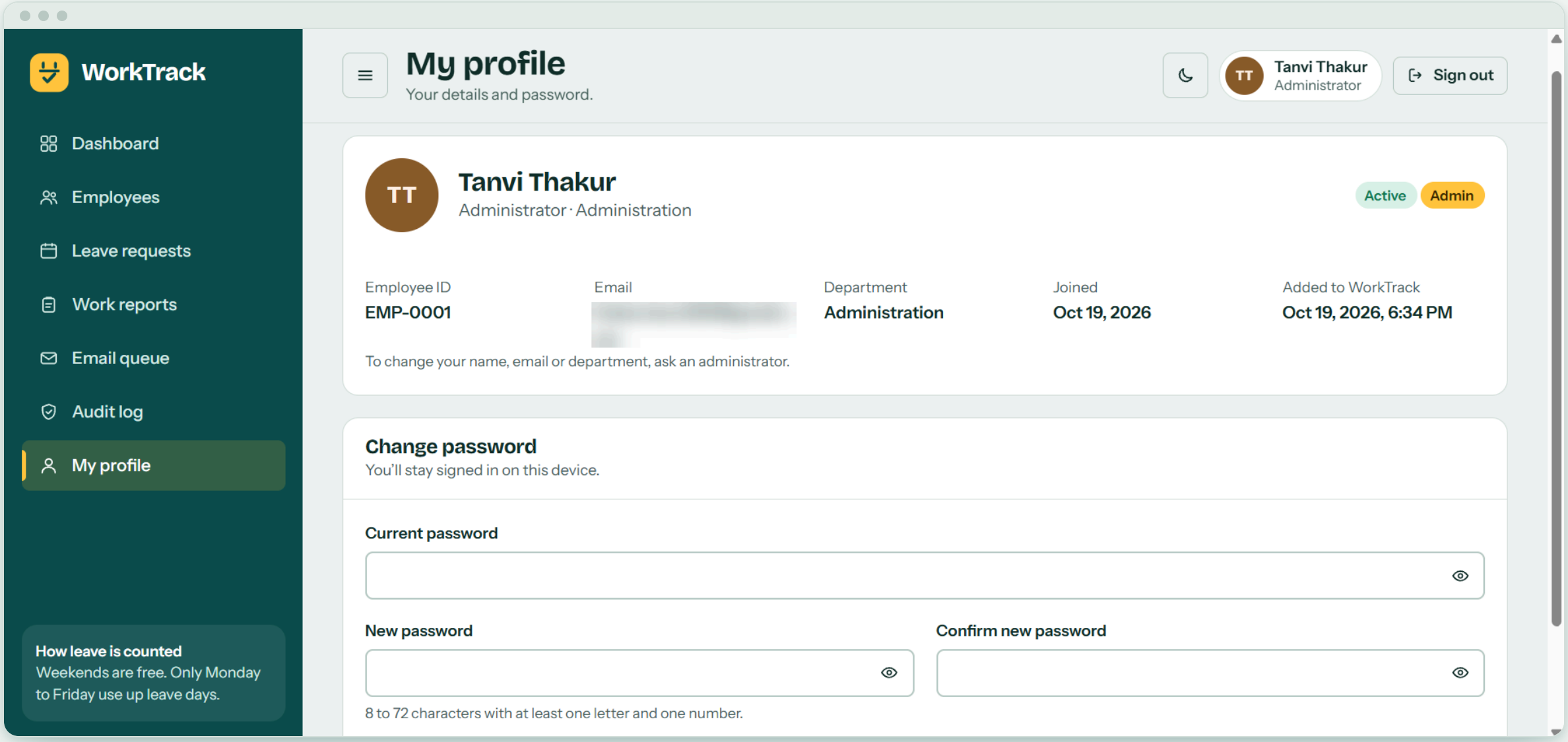





**One interface, two themes.** The admin dashboard in dark mode. The theme toggle in the header switches the whole application.

# Account and security

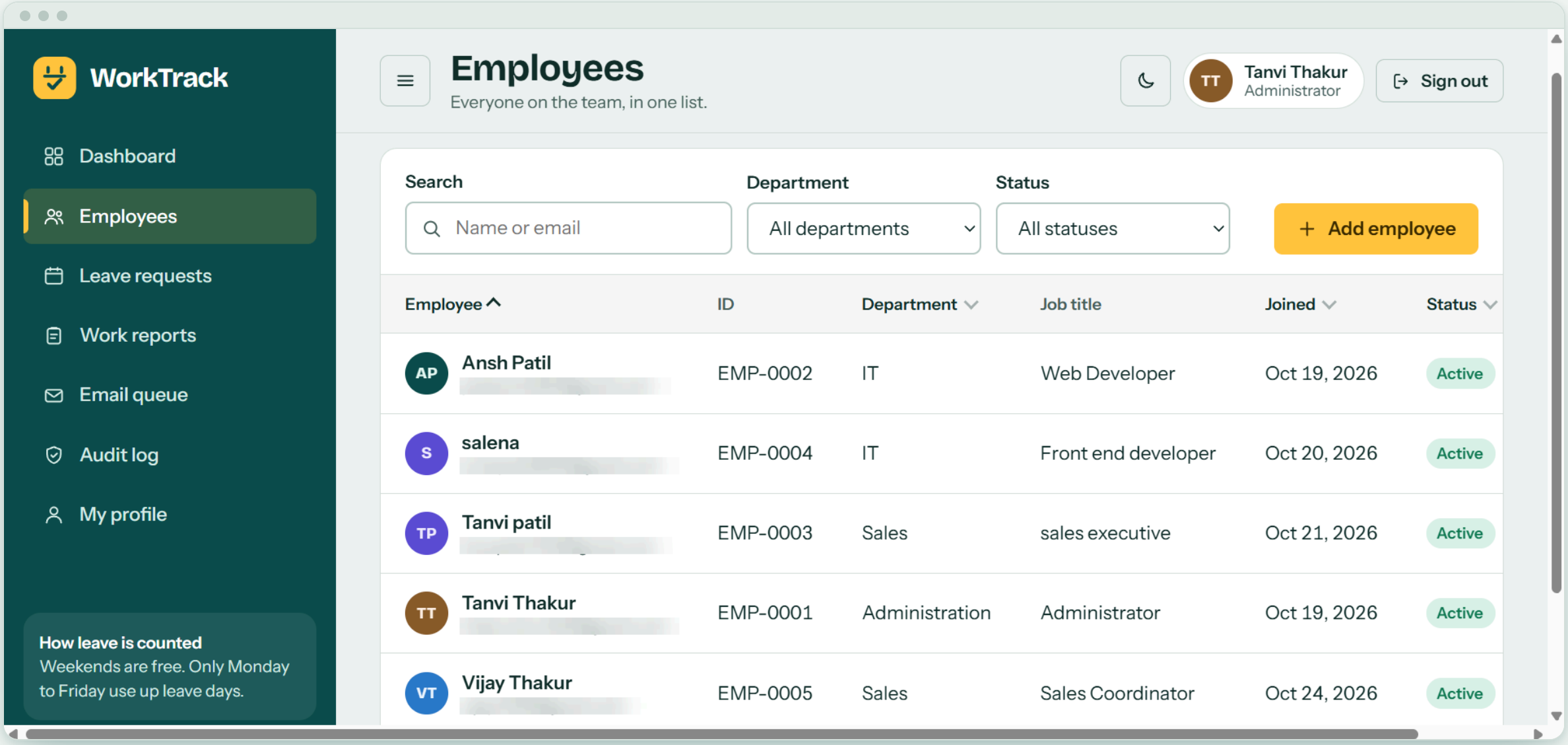





**Users manage their own password.** My profile shows the account details and a change-password form (8 to 72 characters, at least one letter and one number). Email address masked in this portfolio.

# Employee directory

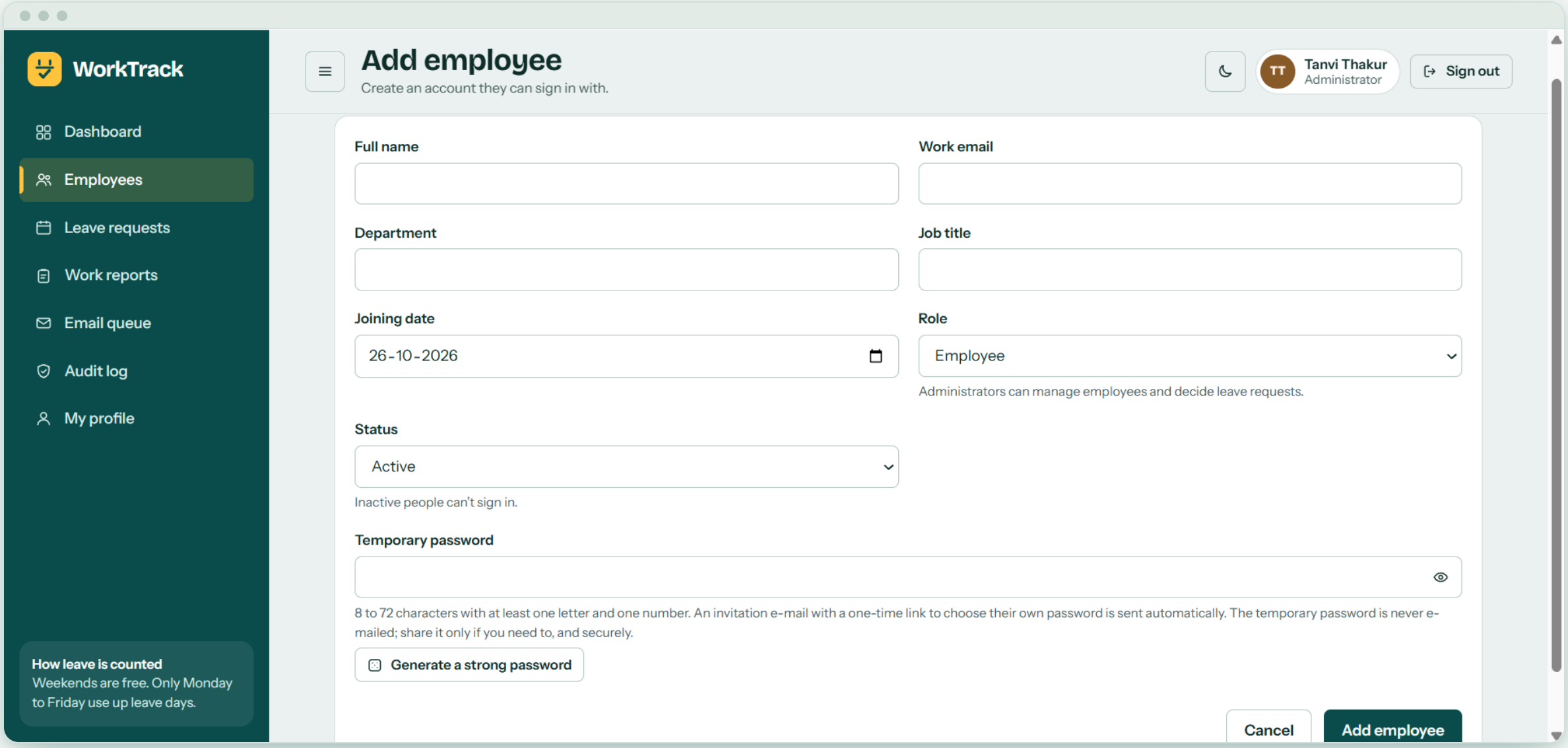





**Find and filter the whole team.** Search by name or email, filter by department and status, and see employee IDs, job titles, joining dates and status. Email addresses masked in this portfolio.

# Adding an employee

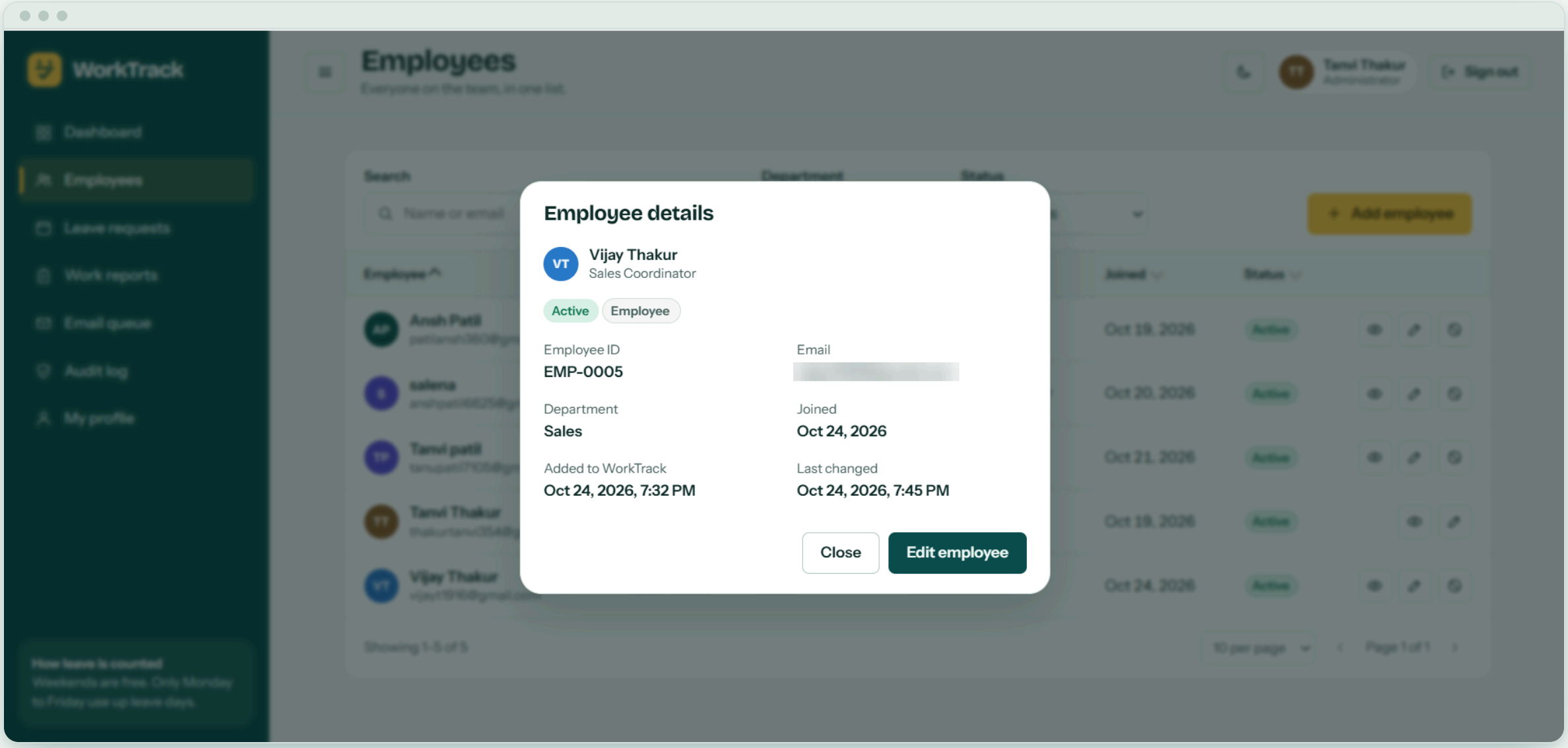





**A guided form for new accounts.** Name, work email, department, job title, joining date, role (Employee or Administrator) and status. A temporary password is set, and an invitation email lets the employee choose their own.

# Employee details

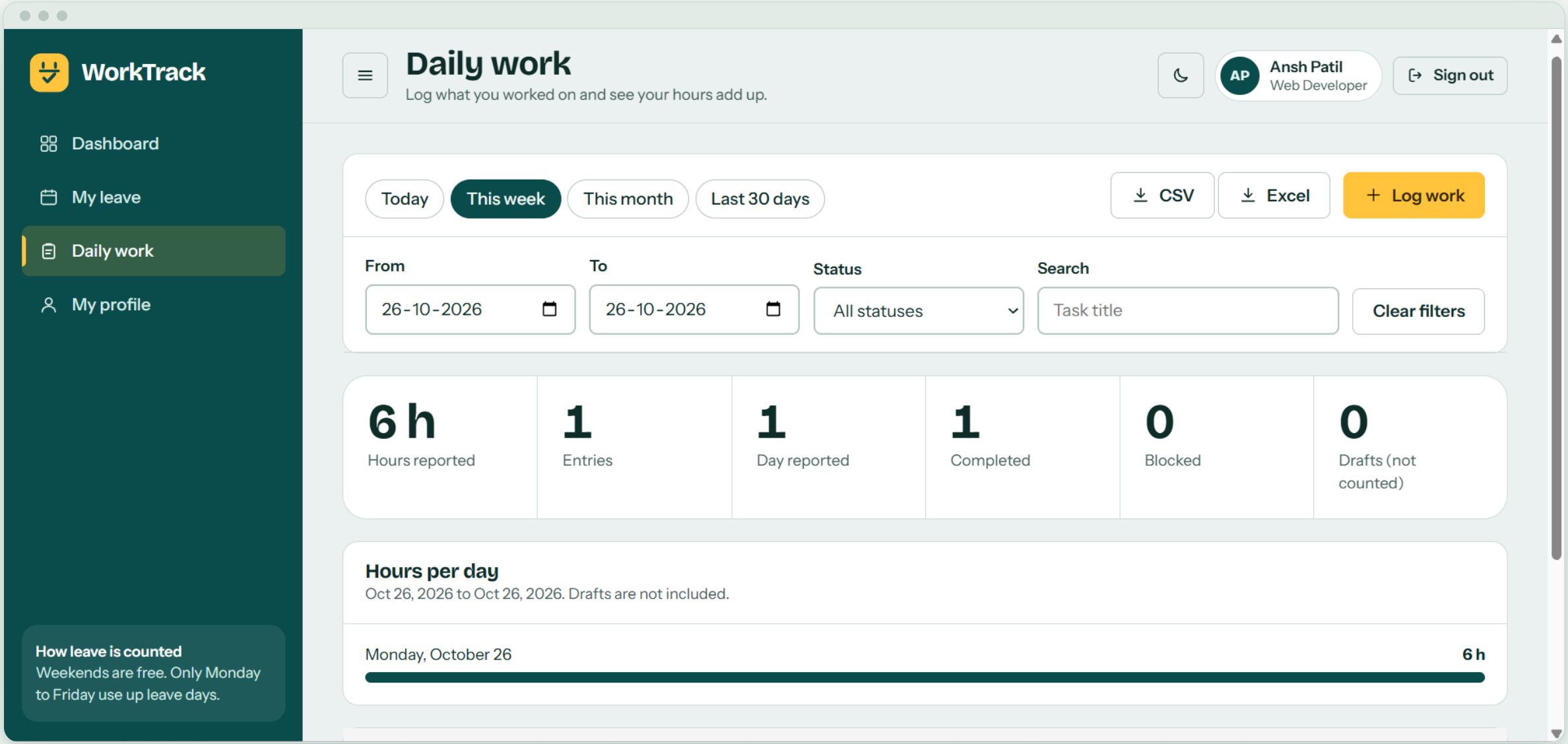





**Record details in one place.** Employee ID, department, joining date, and when the record was added and last changed, with a shortcut to edit. Email address masked in this portfolio.

# Logging daily work

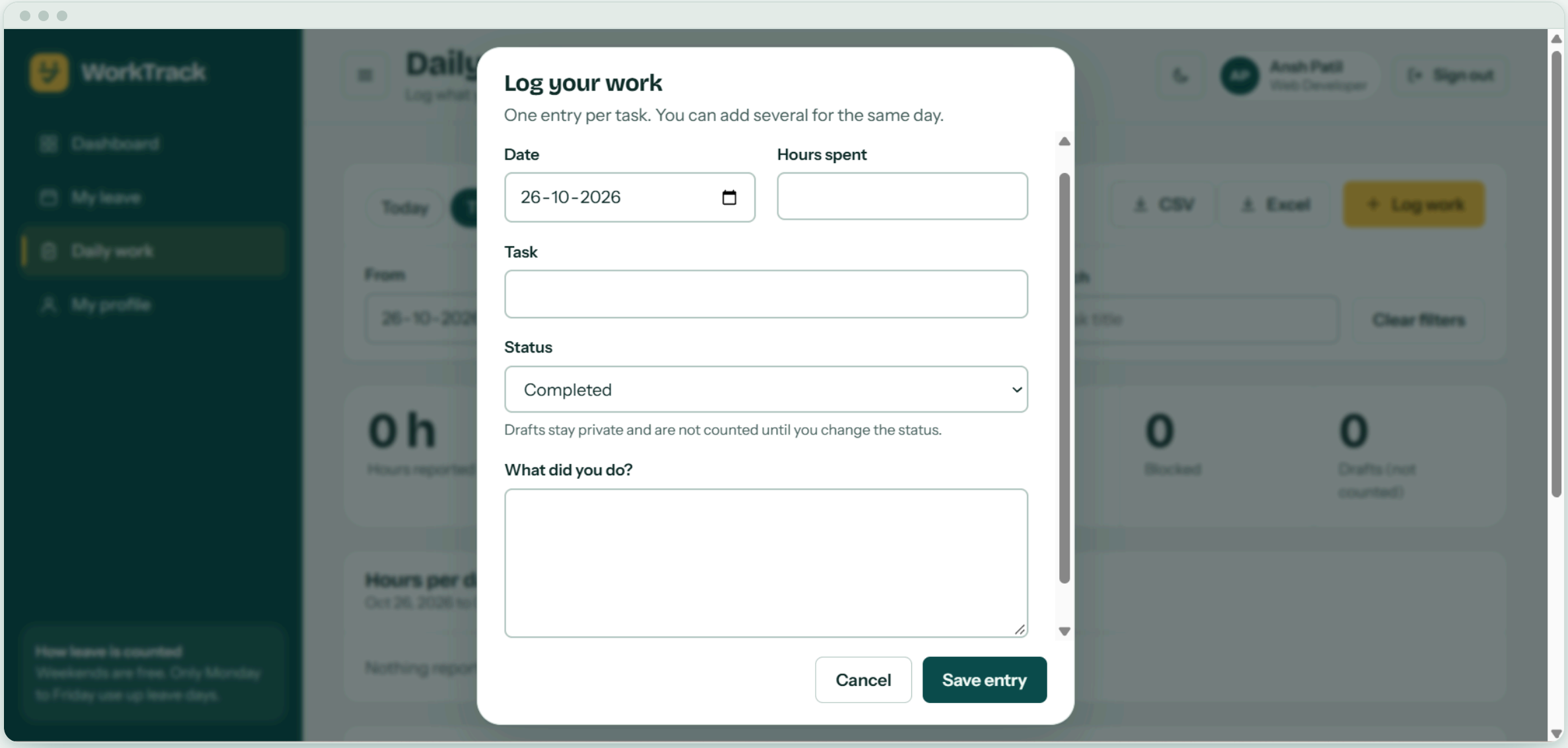





**Employees see their hours add up.** Filter by today, this week, this month or the last 30 days; see hours reported and entries by status; export to CSV or Excel.

# Adding a work entry

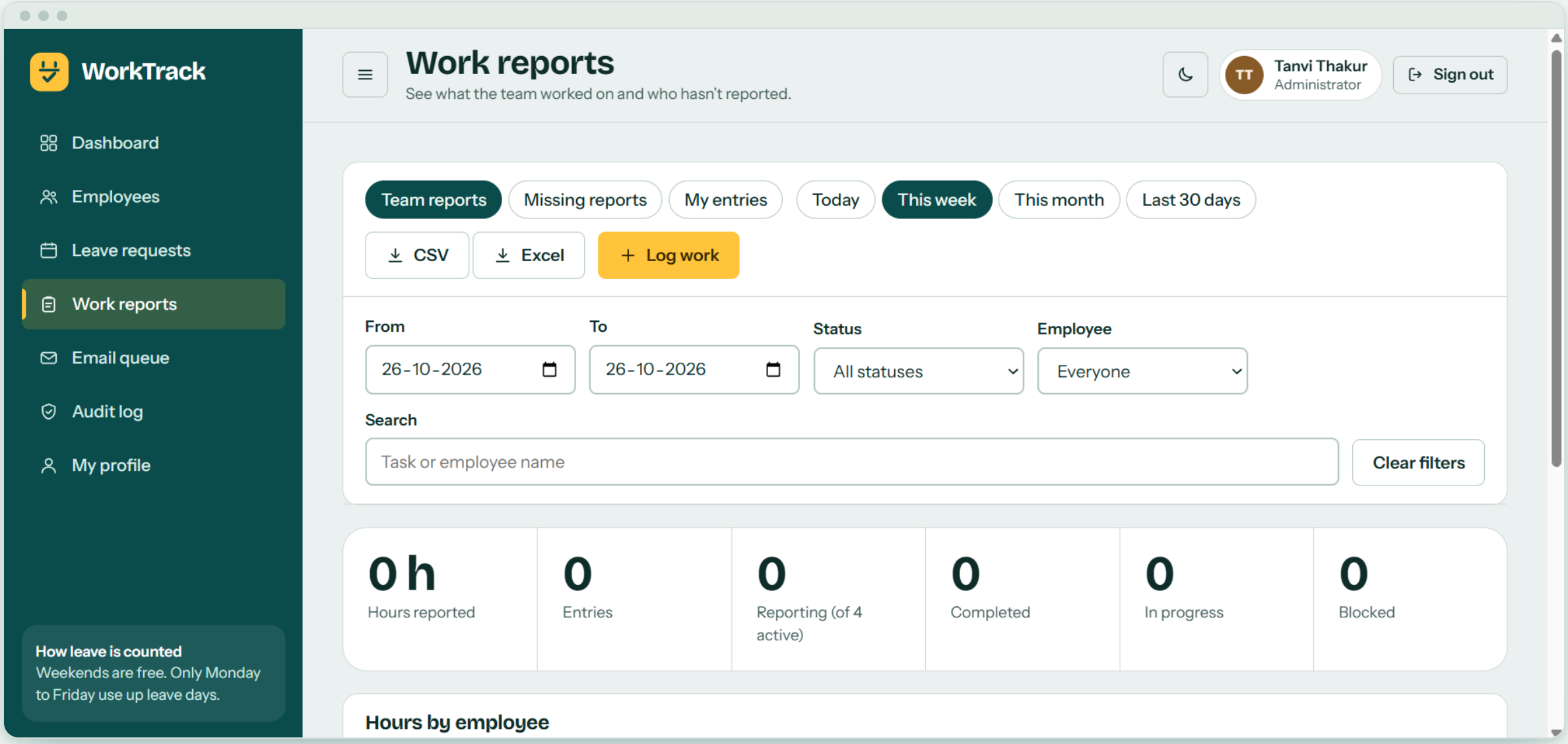





**One entry per task.** Date, hours, task, status and a description of what was done. Drafts stay private and are not counted until the status changes.

# Team work reports

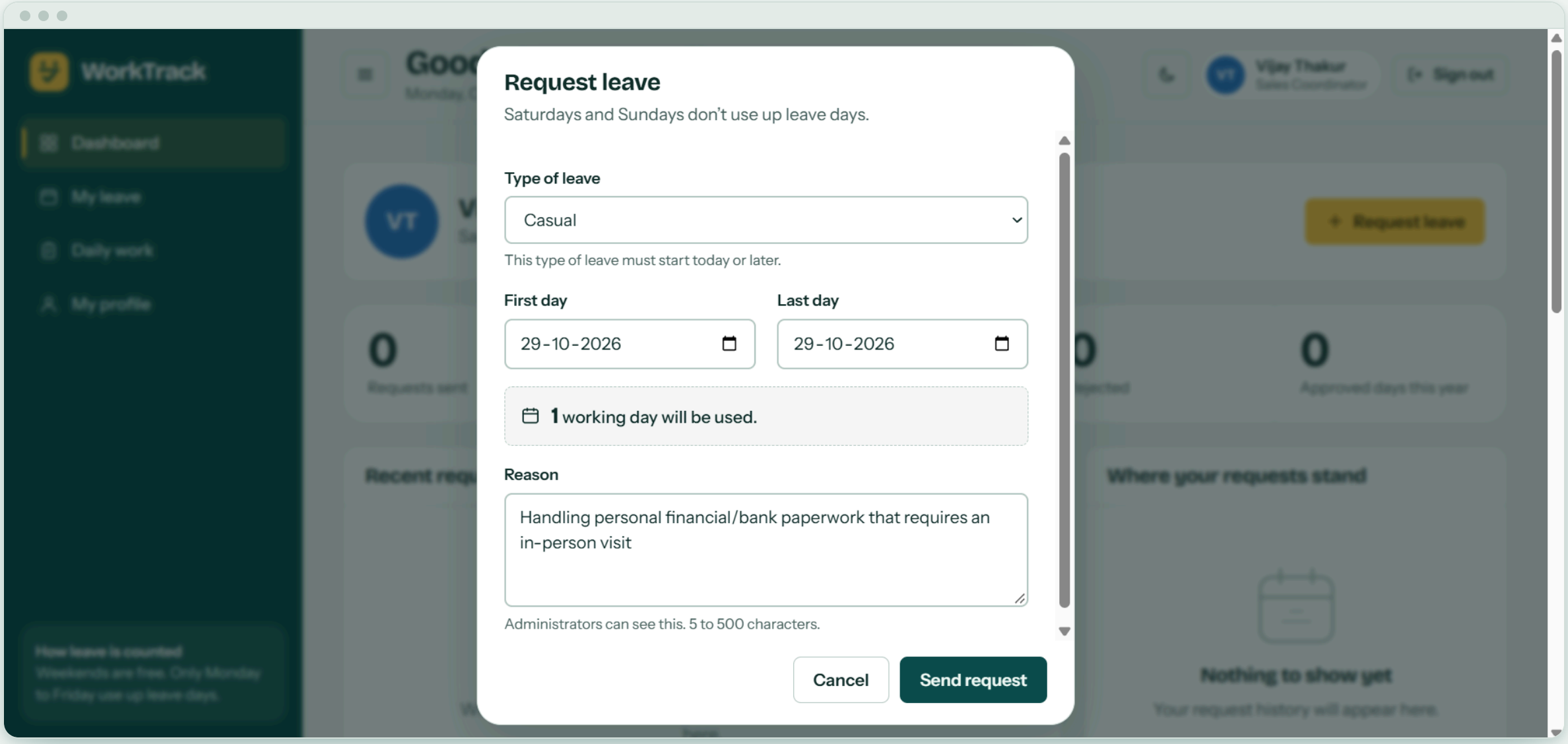





**Administrators review the team's work.** Team reports, missing reports and personal entries, with date presets, status and employee filters, search, and CSV or Excel export. This capture shows a day with no entries.

# Requesting leave

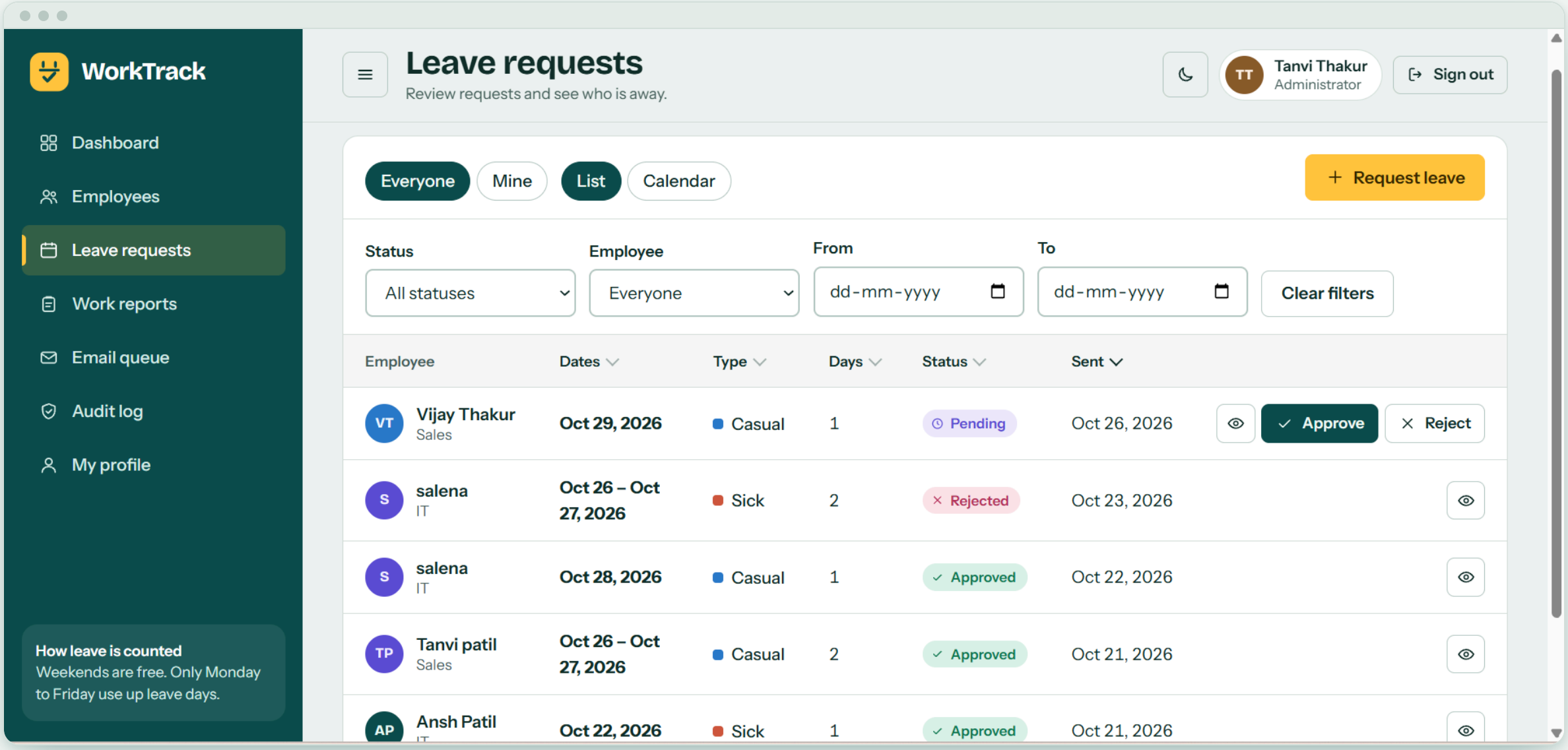





**Clear rules before sending.** Employees choose a type, dates and a reason (5 to 500 characters). The form shows how many working days will be used; weekends do not count.

# Reviewing leave requests

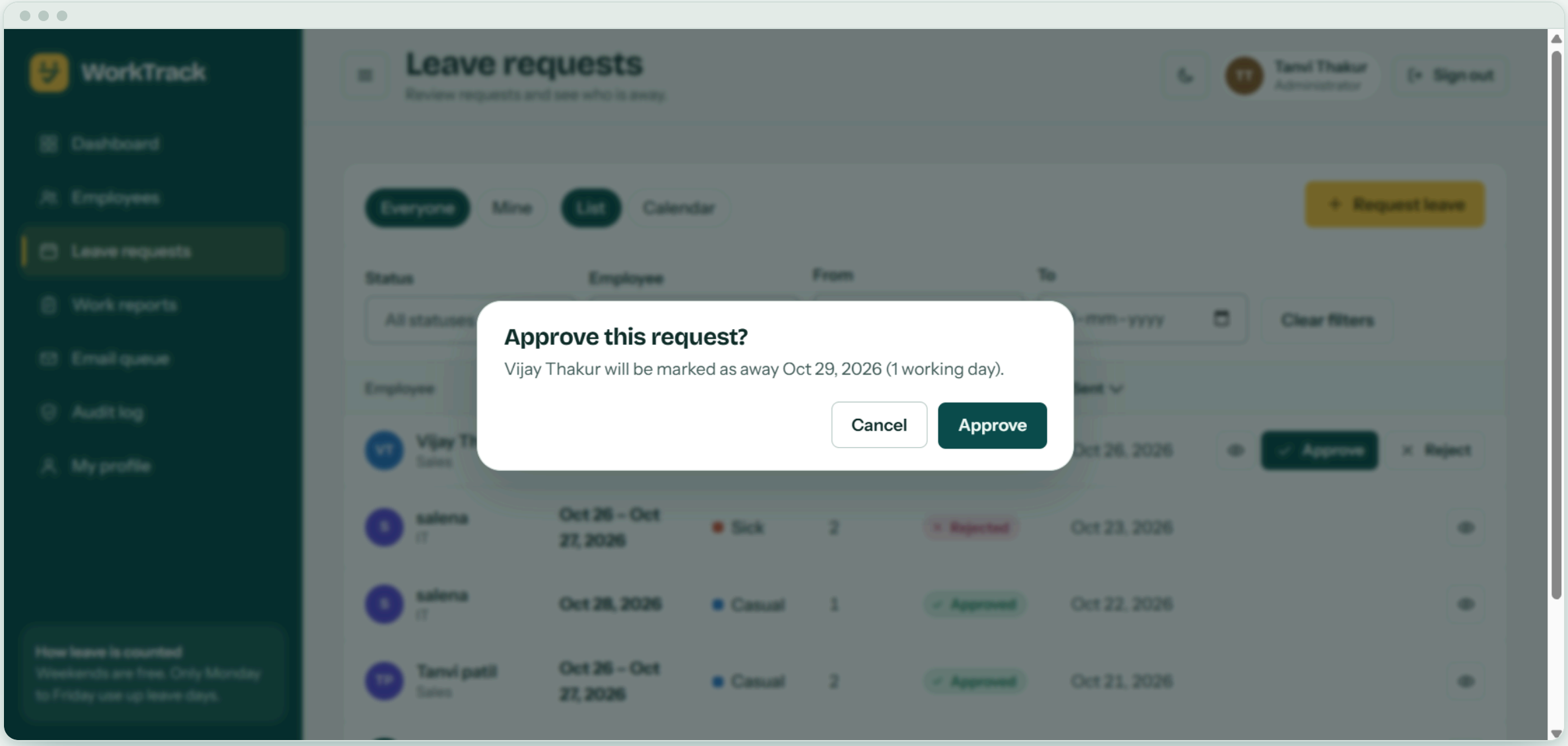





**All requests in one list.** Administrators see everyone's requests or only their own, as a list or calendar, filter by status, employee and dates, and approve or reject pending requests.

# Approving a request

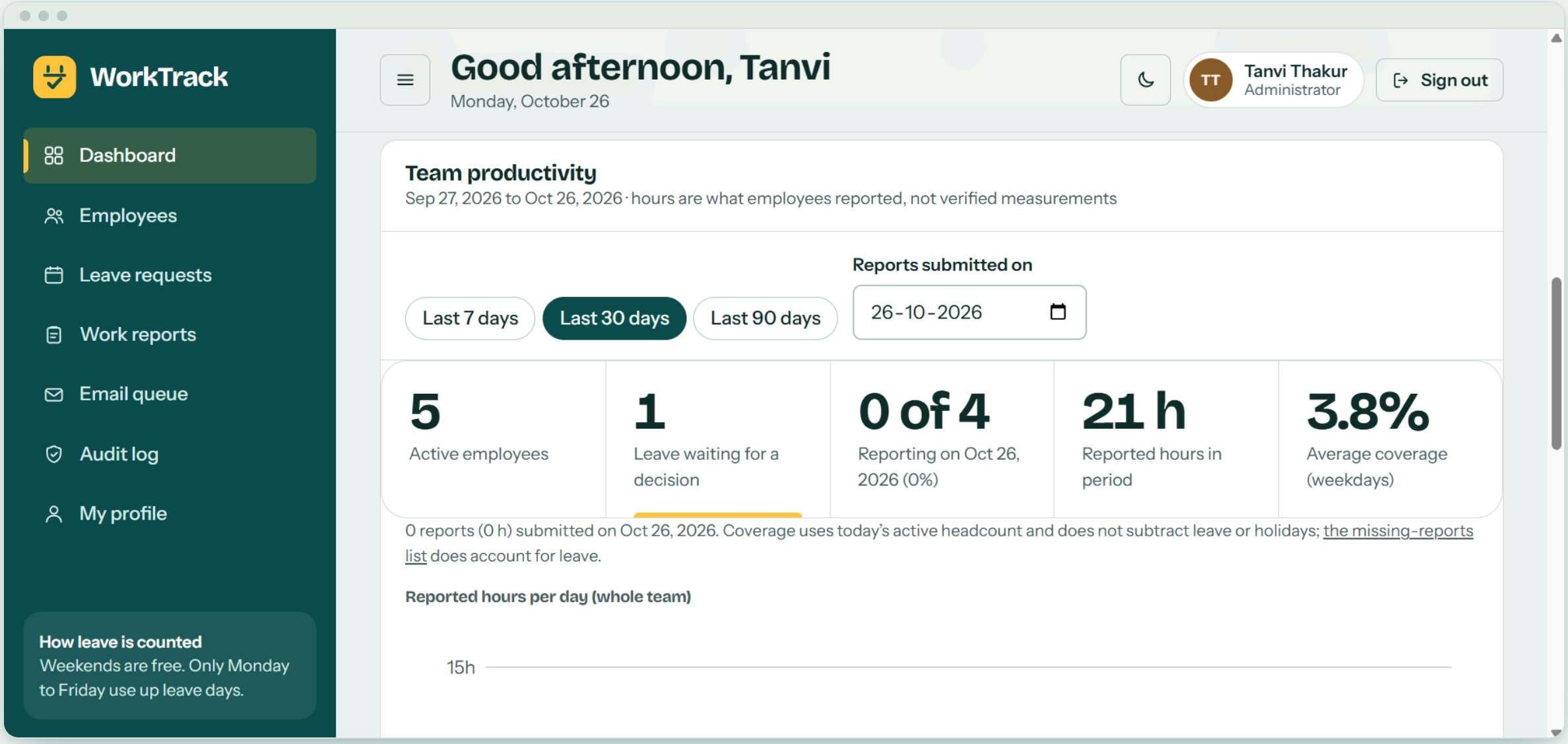





**A confirmation before every decision.** The dialog states who will be marked away and for how many working days before the decision is saved.

# Team productivity

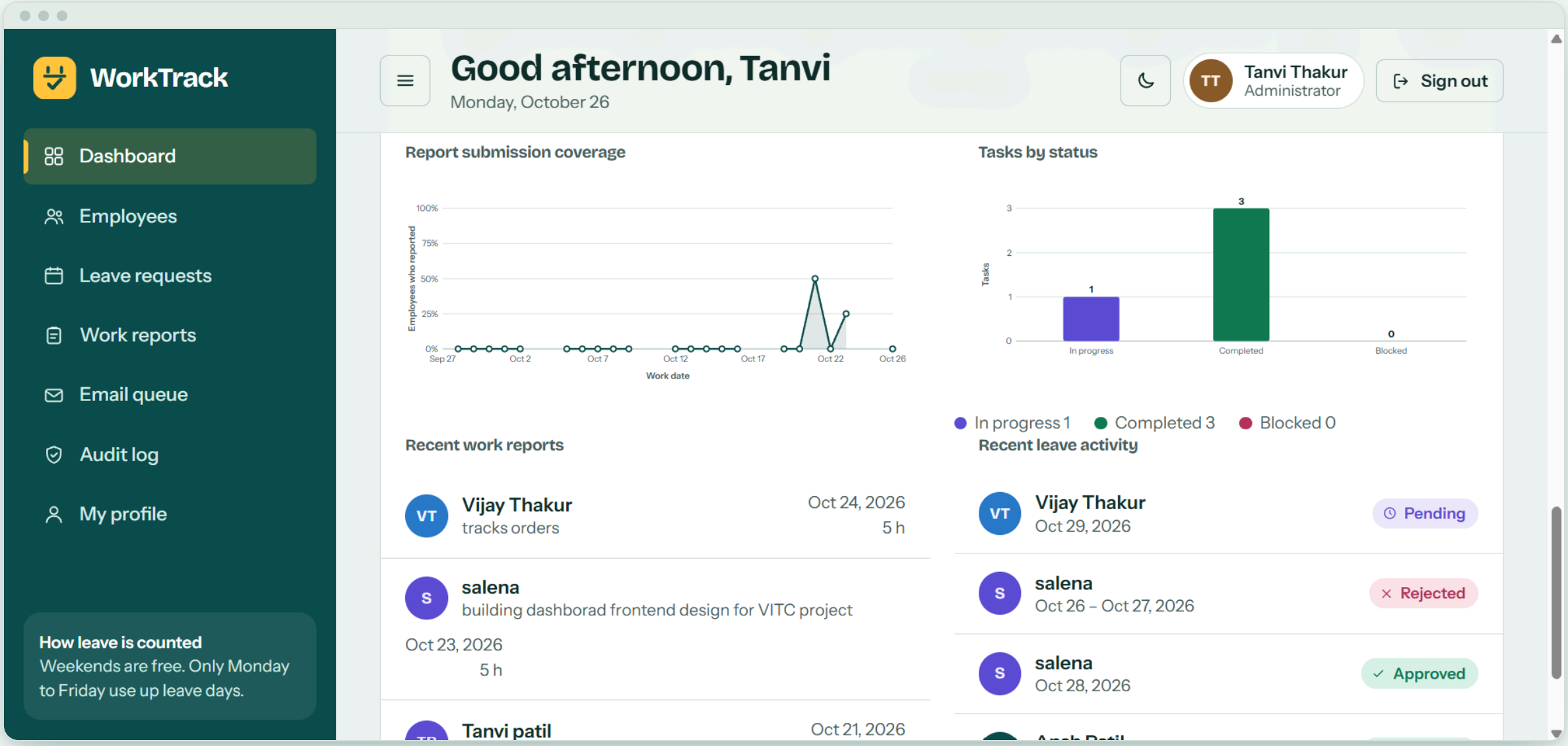





**Summary numbers for a chosen period.** Choose the last 7, 30 or 90 days, or a single day: active employees, leave waiting, reporting coverage, reported hours and average weekday coverage. The page notes that hours are reported by employees, not verified measurements.

# Charts and recent activity

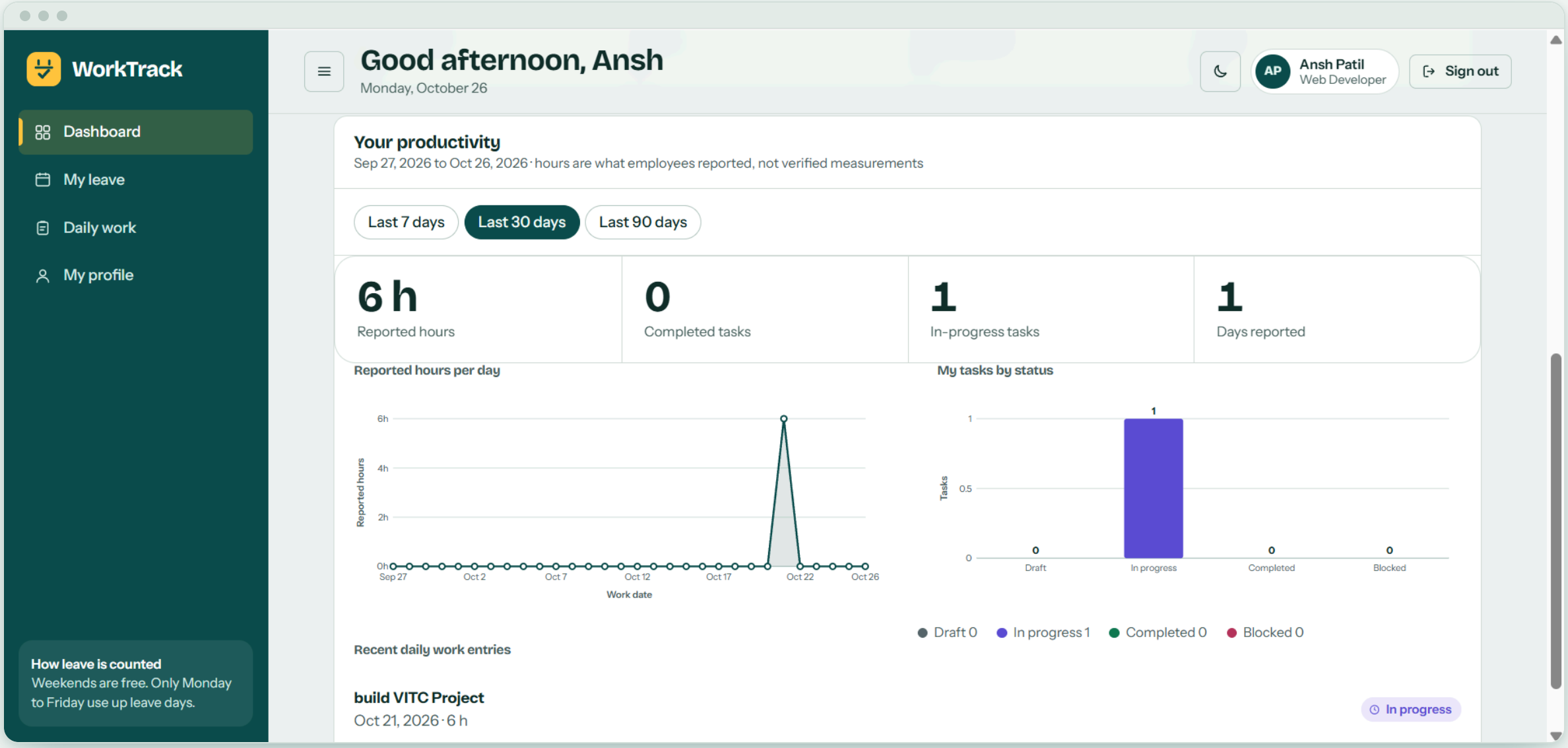





**Trends and recent activity.** Report submission coverage over time, tasks by status, recent work reports and recent leave activity.

# Employee productivity

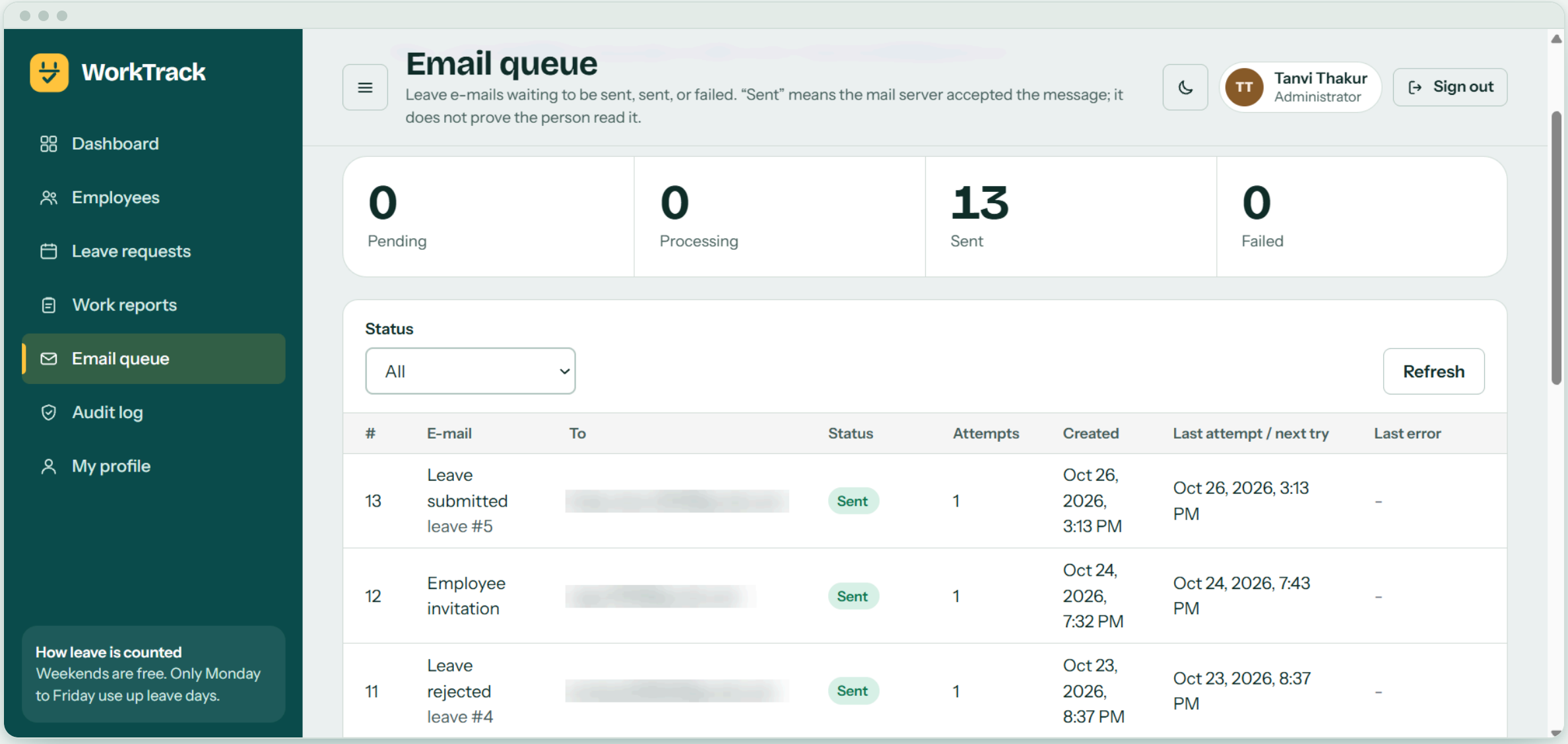





**Each employee's own summary.** Reported hours, completed and in-progress tasks, days reported, hours per day and tasks by status, plus recent daily work entries.

# Email queue

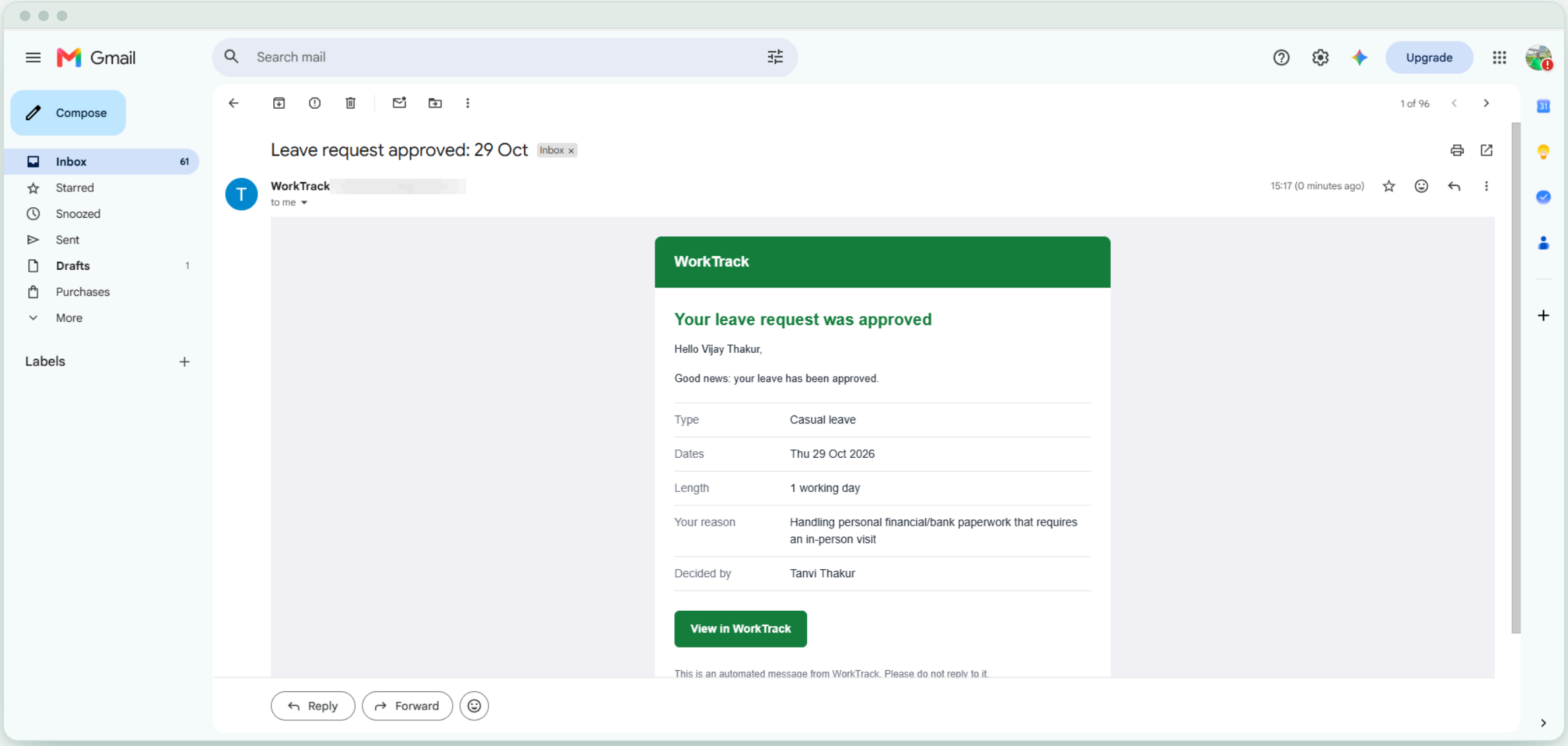





**Notification delivery is visible to admins.** Each message has a status (Pending, Processing, Sent or Failed), attempt count and last error. "Sent" means the mail server accepted the message. Recipient addresses masked.

# Leave decision email

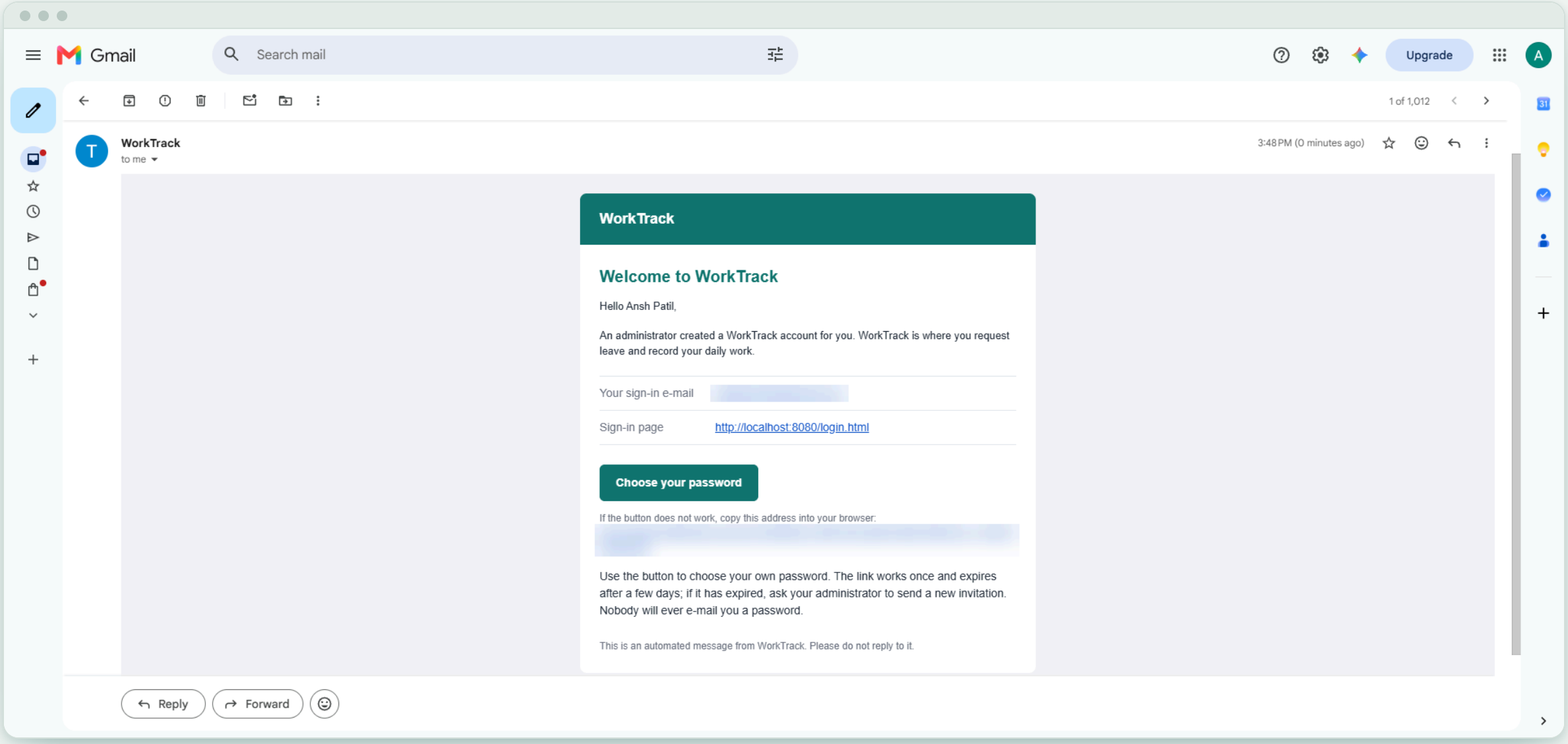





**The employee is told the outcome.** Type, dates, length, the employee's reason and who decided. This is the message as received in a Gmail inbox during testing. Sender address masked.

# Invitation email

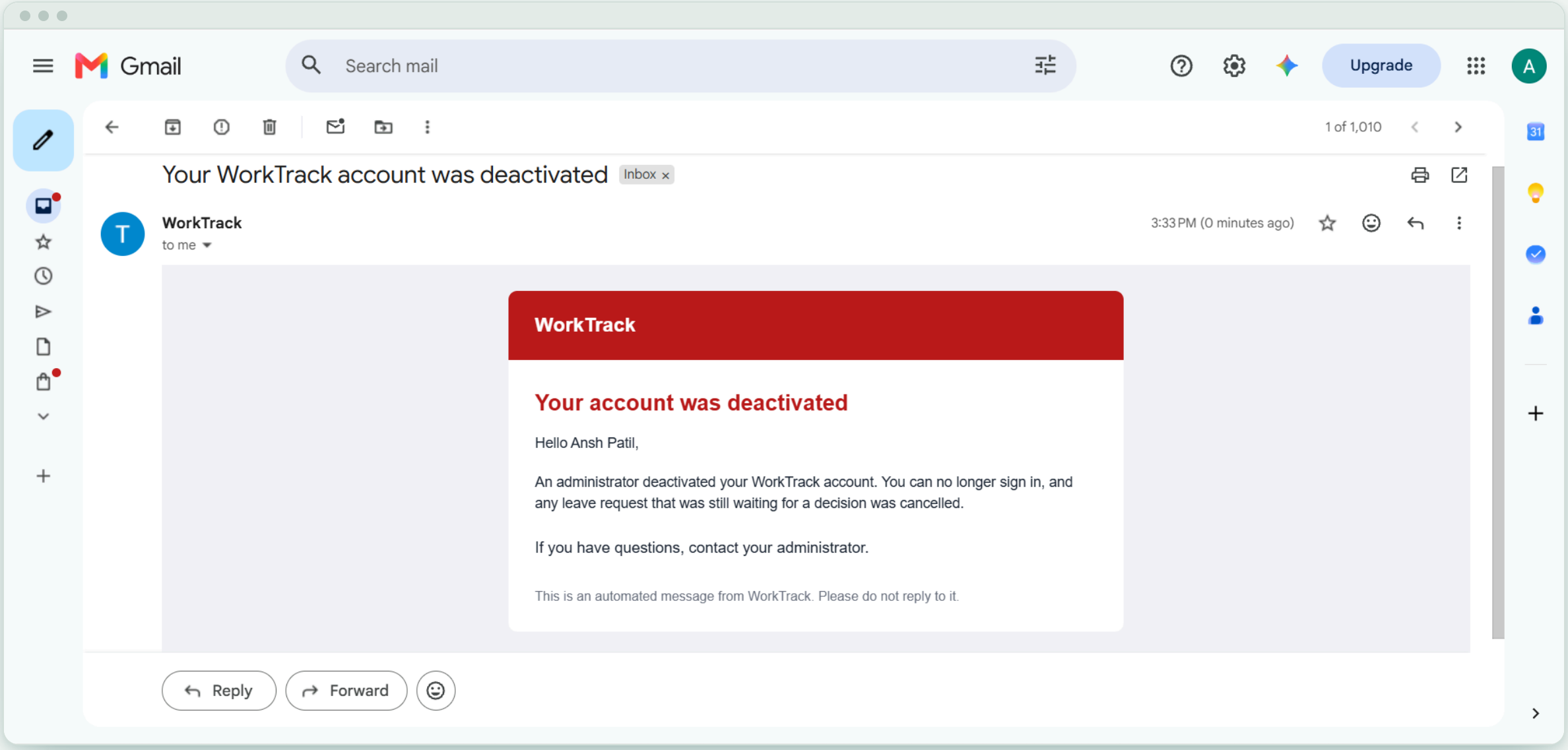





**New employees choose their own password.** A welcome email with a one-time “Choose your password” link; no password is ever sent by email. The sign-in address and the secret link are masked here.

# Account deactivated email

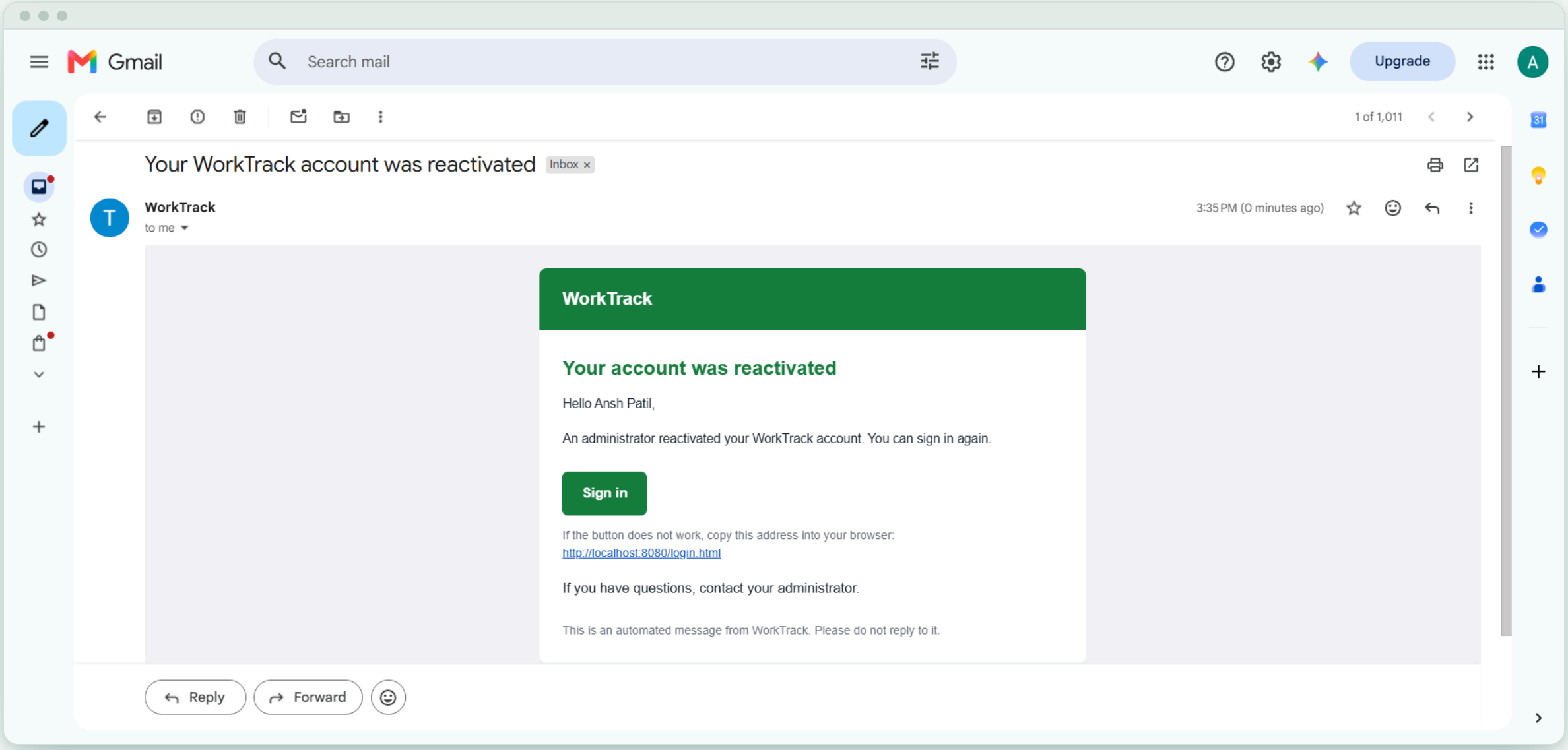





**Account changes are communicated.** When an administrator deactivates an account, the employee is told they can no longer sign in and that any waiting leave request was cancelled.

# Account reactivated email

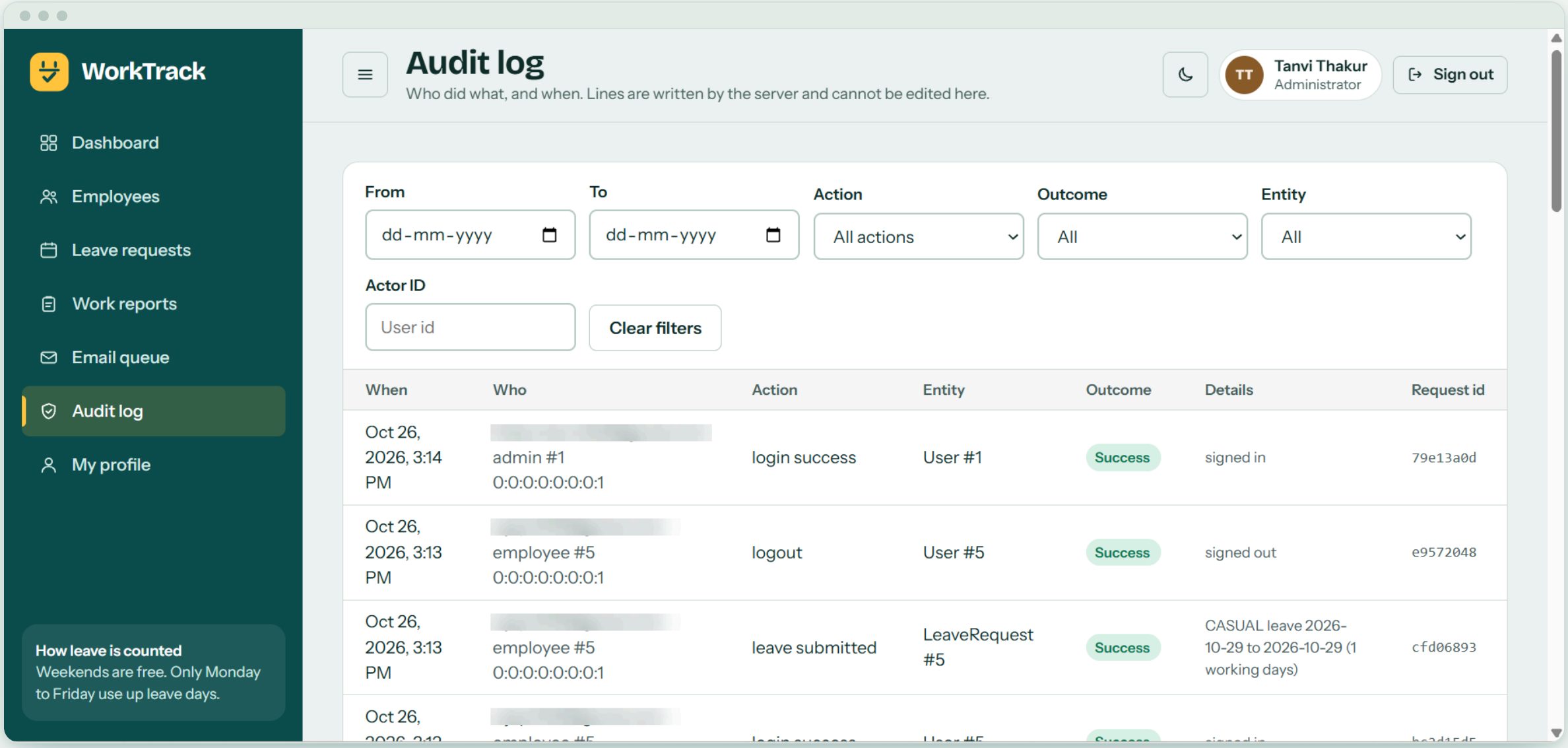





**Access can be restored.** The reactivation email includes a sign-in button. Links point to localhost because the application ran locally.

# Audit log






**Who did what, and when.** Sign-ins, sign-outs and leave submissions with outcome, details and request ID, filterable by date, action, outcome, entity and user. Entries are written by the server and cannot be edited in the app. Addresses masked.

TECHNOLOGY STACK AND ARCHITECTURE

# How it is built

WorkTrack

### Frontend
HTML, CSS and JavaScript pages in the browser

→

### REST APIs
Spring Boot controllers receive requests and return JSON

→

### Business logic
Service layer applies rules such as leave validation and permissions

→

### MySQL
Employees, leave, work entries, notifications and audit records

**FRONTEND**

HTML · CSS · JavaScript

**BACKEND**

Java 17 · Spring Boot · Spring Security · Spring Data JPA

**DATABASE**

MySQL · Flyway migrations

**BUILD AND TEST**

Maven · JUnit

### Authentication layer
Spring Security handles sign-in with a server session, CSRF protection and role checks. Passwords are stored as BCrypt hashes. Permissions are enforced on the server, not only by hiding buttons in the interface.

### Email delivery
Notifications are saved in the database and sent by a background worker over SMTP, with retries and a status for each message. This keeps a mail problem from blocking a leave decision.

KEY FEATURES

# What the screenshots demonstrate

WorkTrack

### Role-based access

Administrator and Employee roles with different pages and permissions.

### Employee directory

Search, filters, employee IDs, details and editing.

### Invitation and password setup

Welcome email with a one-time link; users can change their password.

### Daily work logging

Tasks with hours and status; drafts stay private; CSV and Excel export.

### Leave workflow

Request, confirm, approve or reject, with working days counted automatically.

### Dashboards and charts

Separate views for administrators and employees; light and dark themes.

### Email notifications

Invitations, leave decisions and account changes, with a status queue.

### Audit log

Server-written record of sign-ins and leave and account actions.

PROJECT LEARNINGS

# What I explored and learned



WorkTrack was a way to learn how the parts of a full-stack application fit together. These are the areas I worked through while building it.

### Frontend to backend

Connecting JavaScript pages to Spring Boot REST APIs and handling responses and errors.

### Java and Spring Boot

Organising code into controllers, services and repositories.

### MySQL

Designing tables and relationships, writing queries, and managing schema changes with Flyway.

### Authentication and roles

Sessions, CSRF protection, password hashing and role-based access.

### Business workflows

Employee, daily work and leave workflows, including validation and status changes.

### Reports and email

Dashboard summaries, CSV and Excel export, and a reliable email notification queue.

# Project summary

WorkTrack is a personal full-stack project that manages employees, daily work and leave requests for two roles, administrators and employees. It brings together a JavaScript frontend, a Spring Boot backend, a MySQL database and email notifications.

Building it was a way to explore how frontend, backend, database and business workflows work together.

Java | Spring Boot | Spring Security | MySQL | JavaScript | HTML | CSS | Maven

The screenshots were taken from the application running locally; it is not publicly deployed. Email addresses and the invitation link are masked.